

IEORE4004

Optimization Models and Methods

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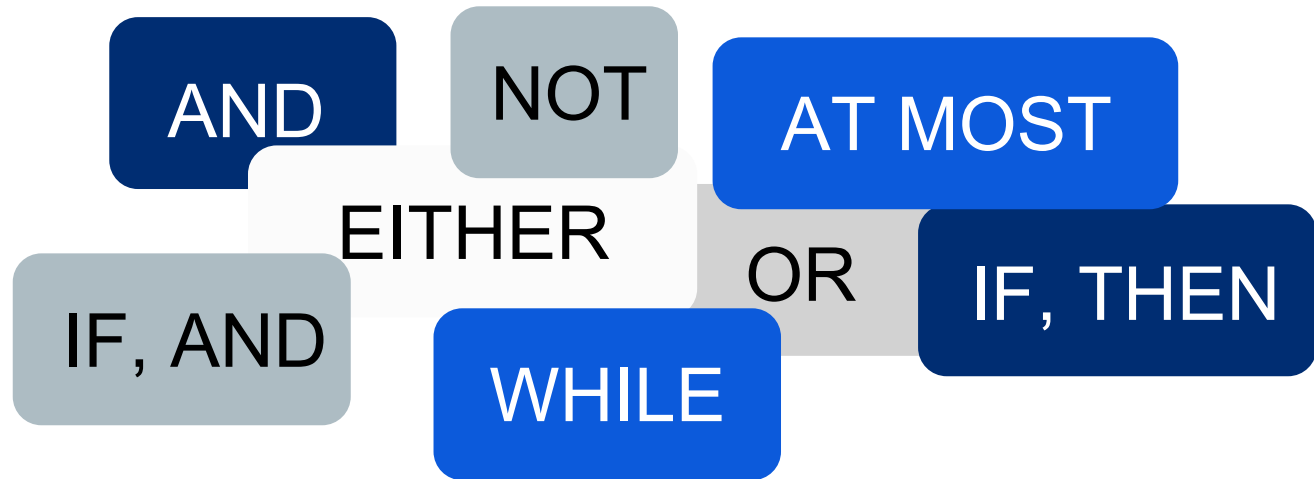
Outline

- Modeling Logical Constraints
- Modeling Piecewise Linear Functions

Modeling Logical Conditions

One of the many powerful uses of binary variables is in the modeling of *logical* conditions

Some logical conditions:



Let's illustrate the logical features with a ***capital budgeting*** model

Example | Capital Budgeting

Suppose a company is making allocation decisions on R&D investments, and there are 6 internal opportunities to consider:

Investment	1	2	3	4	5	6
Investment Cost (in K)	\$5.50	\$6.50	\$7.00	\$9.00	\$10.50	\$13.00
NetPresentValue (in K)	\$15.00	\$19.00	\$19.00	\$25.00	\$29.00	\$34.00

- Either an entire project investment is made, or not at all
- No investment can be made twice
- The present budget is \$23K and all unused portions of budget count toward total NPV

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- No investment can be made twice
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What type of a problem is this?
What's the optimal investment plan?

Example | Capital Budgeting

For this problem, we consider:

- **all or nothing** capital budgeting decisions
- no investment can be made twice
- a budget cap exists on the maximum amount to invest
- each investment:
 - has a certain cost
 - returns a certain value

Capital Budgeting Example

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- **all or nothing** decisions,
- no investment can be made twice,
- a budget cap exists,
- each investment:
 - has a certain cost
 - returns a certain value

Knapsack Problem

$$\text{Maximize } \sum_{j=1}^n v_j x_j$$

$$\text{subject to } \sum_{j=1}^n w_j x_j \leq w_0$$

$$x_j \in \{0, 1\}, \quad j = 1, 2, \dots, n$$

Capital Budgeting Example

Then the IP becomes:

$$\text{Maximize } 9.5 x_1 + 12.5 x_2 + 12 x_3 + 16 x_4 + 18.5 x_5 + 21 x_6$$

$$\text{S.t. } 5.5 x_1 + 6.5 x_2 + 7 x_3 + 9 x_4 + 10.5 x_5 + 13 x_6 \leq 23$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \in \{0,1\}$$

Capital Budgeting Example

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$$\begin{aligned} \text{Maximize} \quad & 9.5 x_1 + 12.5 x_2 + 12 x_3 + 16 x_4 + 18.5 x_5 + 21 x_6 \\ \text{S.t.} \quad & 5.5 x_1 + 6.5 x_2 + 7 x_3 + 9 x_4 + 10.5 x_5 + 13 x_6 \leq 23 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \in \{0,1\} \end{aligned}$$

While I am solving this IP using Gurobi, try it yourselves!

Capital Budgeting Example

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While I am solving this IP using Gurobi, try it yourselves!



What is the optimal investment strategy?

Capital Budgeting Example

The optimal solution is to undertake **investments 1, 2, and 5**

- This uses \$22.5K of the \$23K available resources (i.e., within budget)
- The associated NPV with this decision is \$63K (\$15K + \$19K + \$29K)
- Plus \$.5K left over from unused budget, so total of **\$63.5K**

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! There is also another optimal solution!

- Undertake **investments 2, 3, and 4**

In Gurobi, we can find the other optimal solutions for an IP by taking an iterative approach – *might be time consuming for larger problems!*

Capital Budgeting: Example & Logical Conditions

Now suppose the company faces one of the following (mutually exclusive) events:

-  Invest in **either** investment 2 or 3, but **not both**
-  Investment in 4 **requires** investment in 6, and **vice versa**
-  At most **one** of investments 2, 4, **or** 5 can be invested in
-  Exactly **two** investments must be invested in
-  Investment in 2, **requires** investments in 4 **and** 6

Choosing Between Investments

0 stands for **False**

1 stands for **True**

When an **event takes place** we will represent it with **True** and when it is **not taking place** we will represent with **False**.

We were told that we can invest in only one Investment, 2 or 3, at once

So, **either** Investment 2 or Investment 3 can be True at the same time

Investment 2	Investment 3
True (1)	False (0)
False (0)	True (1)

Event 1: Can do 2 or 3, *but not both*

One tool to assist with modeling logical conditions is a ***truth table***

- List out all possible combinations of binary variable values:

Investment 2	Investment 3
0	0
0	1
1	0
1	1

- What's the equivalent constraint for this event?

Event 1: Can do 2 or 3, *but not both*

One tool to assist with modeling logical conditions is a **truth table**

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0	0
0	1
1	0
1	1

- What's the equivalent constraint for this event?

$$x_2 + x_3 \leq 1$$

Because there are two optimal solutions to the problem, the solution to undertake 1, 2, and 5 is still possible, and so this condition **does not impact** the optimal objective function value

Event 2: If 4, then 6... and vice versa

List out all possible combinations of binary variable values:

Investment 4	Investment 6
0	0
0	1
1	0
1	1

○ What's the corresponding constraint for this event?

Event 2: If 4, then 6... and vice versa

List out all possible combinations of binary variable values:

Investment 4	Investment 6
0	0
0	1
1	0
1	1

○ What's the corresponding constraint for this event?

$$x_4 - x_6 = 0$$

The original optimal solution of undertaking 1, 2, and 5 is still optimal, but the alternate optimal solution of undertaking 2, 3, and 4 will not work.

Event 3: At most one of 2, 4, and 5

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

Investment 2	Investment 4	Investment 5
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Event 3: At most one of 2, 4, and 5

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

$$x_2 + x_4 + x_5 \leq 1$$

Investment 2	Investment 4	Investment 5
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

- No longer will any of the original optimal solutions work
- The new optimal solution is to undertake investments 1, 3, and 5
 - This leads to a new optimal objective function value of \$63K

Event 4: Exactly 2 of 6 Investments

This one may be more obvious than previous ones...

- What's the corresponding constraint?

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- What's the corresponding constraint?

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 2$$

The new optimal solution is to undertake investments 4 and 6

- This leads to a new optimal objective function value of (only) \$60K

Event 5: If 2, then 4 and 6

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

Investment 2	Investment 4	Investment 6
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Event 5: If 2, then 4 and 6

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

Investment 2	Investment 4	Investment 6
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

There are actually two ways to model this!

1. Model with two constraints:

$$x_4 - x_2 \geq 0 \text{ and } x_6 - x_2 \geq 0$$

2. A single constraint:

$$0.5x_4 + 0.5x_6 - x_2 \geq 0$$

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Which is better?

- It turns out that the 1st is *stronger* than the 2nd
 - Provides more information
- Can derive second from the 1st, *but not first from 2nd*

Integer Programming Examples

Question: If we want to restrict x to be one of the elements $\{2, 7, 12\}$, how should we model it?

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Integer Programming Examples

Question: How can we restrict x to be either 0 or between particular positive bounds?

In algebraic notation: $x = 0 \text{ or } l \leq x \leq u$

Integer Programming Examples

Question: If we want to have either one of these constraints:

$2x_1 + 3x_2 \geq 14$ **or** $5x_2 - 7x_3 \leq 3$ how should we model it?

Modeling Fixed & Piecewise Costs

Modeling Fixed Costs

In a typical production planning problem involving N products, the production cost for product j may consist of:

- a fixed cost d_j independent of the amount produced, and
- a variable cost c_j per unit

Thus, if x_j is the production level of product j , its production cost function may be written as:

$$f_j(x_j) = \begin{cases} d_j + c_j x_j, & x_j > 0 \\ 0, & x_j = 0 \end{cases}$$

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And the objective will become: $\text{Min } Z = \sum_{j \in N} f_j(x_j)$

Modeling Fixed Costs

If it is known that $0 \leq x_j \leq u_j$ and $d_j > 0$ then we can define a binary variable y_j that indicates when the fixed cost is incurred, so that:

$$\begin{aligned} y_j &= 1 \text{ if } x_j > 0 \\ y_j &= 0 \text{ if } x_j = 0 \end{aligned}$$

Then the contribution to cost due to x_j may be written as:

$$f_j(x_j) = d_j y_j + c_j x_j$$

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$$f_j(x_j) = d_j y_j + c_j x_j$$

with the constraints:

$$\begin{aligned} x_j &\leq u_j y_j, \\ x_j &\geq \epsilon y_j, \end{aligned} \quad \text{where } \epsilon \text{ is a very small number}$$
$$\begin{aligned} x_j &\geq 0, \\ y &= 0 \text{ or } 1. \end{aligned}$$

Example: Mobile Care Foundation

MCF aims to control asthma in schoolchildren from low-income families in the greater Chicago region through asthma vans that visit elementary schools on weekdays.



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- Operations involve a complex hierarchy of resource allocation decisions aimed at maximizing MCF's effectiveness
- Now due to funding shortfall, the management is forced to review the performance metrics and execute improvement initiatives.

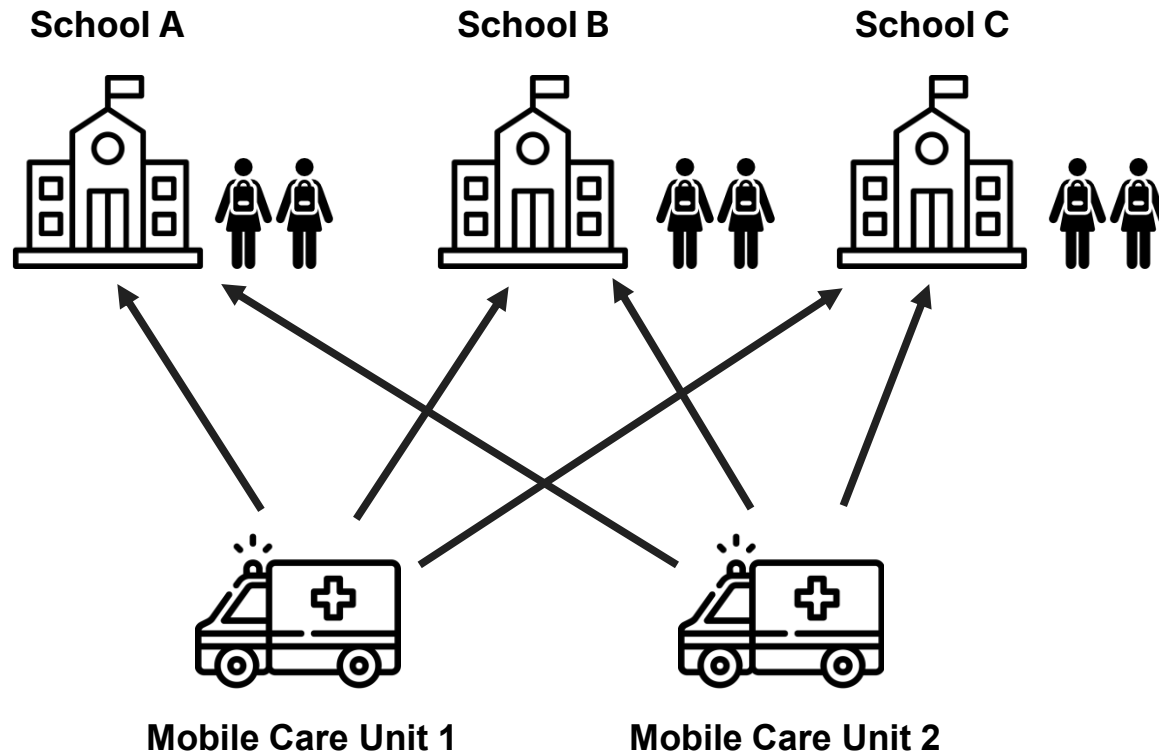


MCF Optimization

After scaling down, MCF now has 2 vans and 3 schools to cover.

You want to tell the MCF planners which van should go to which school and how many appointments they should cover for one day.

Note that at least 60% of the demand must be covered at each school.



	Van 1	Van 2
Fixed cost	\$40	\$60
Appointment cost	\$5	\$8
Capacity (appts.)	30	80

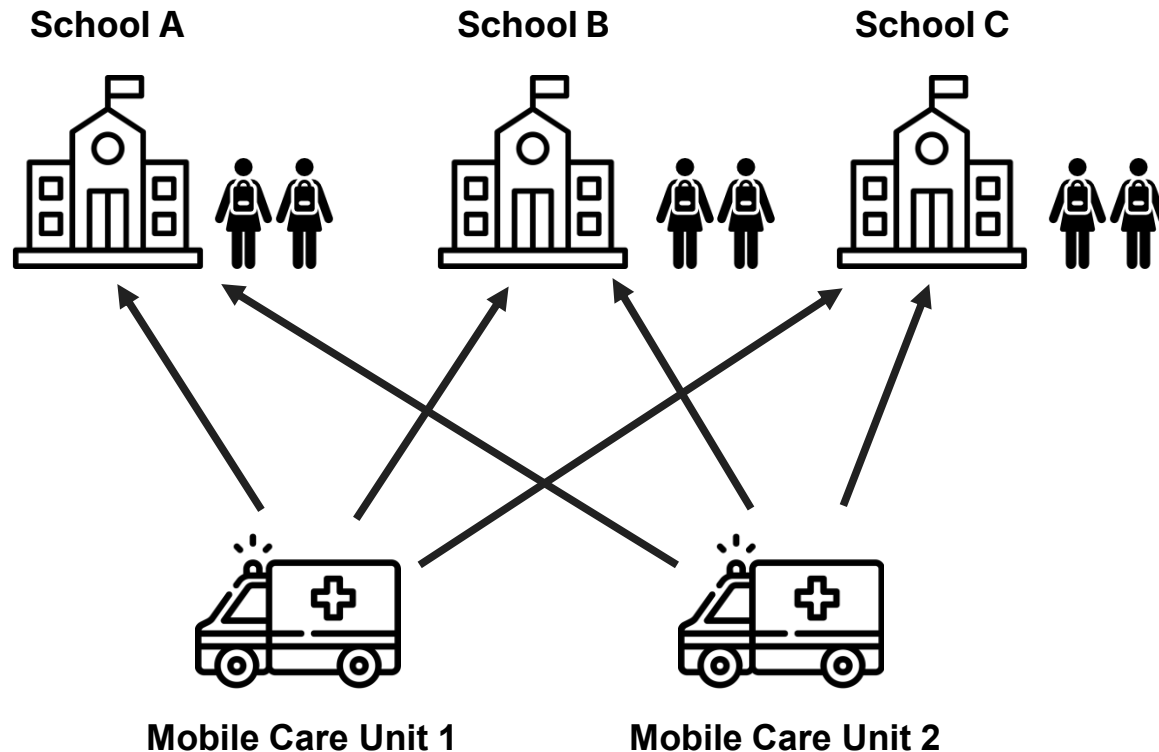
School	Demand for Appointments
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B	50
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? With given information, how would you assign MC units to schools?

MCF Resource Matching

Sets

- Set of schools $S \in \{A, B, C\}$
- Set of vans $V \in \{1, 2\}$

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- Set of vans $V \in \{1, 2\}$

Parameters

- Number of appointments for each school $(a_{S_A}, a_{S_B}, a_{S_C})$
- Fixed cost for each van (f_{V_1}, f_{V_2})
- Appointment cost for each van (g_{V_1}, g_{V_2})
- Capacity for each van (c_{V_1}, c_{V_2})

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- Capacity for each van (c_{V_1}, c_{V_2})

Variables

- $x_{[school, van]}$: Number of appointments assigned to a school by a van
 $(x_{A,1}, x_{A,2}, x_{B,1}, x_{B,2}, x_{C,1}, x_{C,2})$
- $y_{[school, van]}$: if a school is assigned to a van, 1; otherwise, 0
 $(y_{A,1}, y_{A,2}, y_{B,1}, y_{B,2}, y_{C,1}, y_{C,2})$

MCF Resource Matching

Objective

MCF Resource Matching

Objective



1. Minimize the assignment cost

Total Cost = Fixed cost + Variable appointment cost

- Fixed: $40 \cdot (y_{A,1} + y_{B,1} + y_{C,1}) + 60 \cdot (y_{A,2} + y_{B,2} + y_{C,2})$
- Appt.: $5 \cdot (x_{A,1} + x_{B,1} + x_{C,1}) + 8 \cdot (x_{A,2} + x_{B,2} + x_{C,2})$

MCF Resource Matching

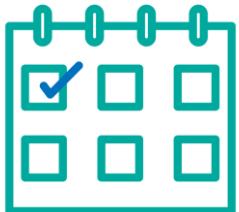
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- Appt.: $5 \cdot (x_{A,1} + x_{B,1} + x_{C,1}) + 8 \cdot (x_{A,2} + x_{B,2} + x_{C,2})$



2. Maximize the number of appointments

Number of appointments:

- $x_{A,1} + x_{B,1} + x_{C,1} + x_{A,2} + x_{B,2} + x_{C,2}$

MCF Resource Matching

Constraints

MCF Resource Matching

Constraints

1. Cover at least 60% of the demand for appointments at each school

- School 1: $x_{A,1} + x_{A,2} \geq 40 \cdot 60\%$
- School 2: $x_{B,1} + x_{B,2} \geq 50 \cdot 60\%$
- School 3: $x_{C,1} + x_{C,2} \geq 35 \cdot 60\%$

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2. Capacity of a van cannot be exceeded

- Van 1: $x_{A,1} + x_{B,1} + x_{C,1} \leq 30$
- Van 2: $x_{A,2} + x_{A,2} + x_{C,2} \leq 80$

MCF Resource Matching

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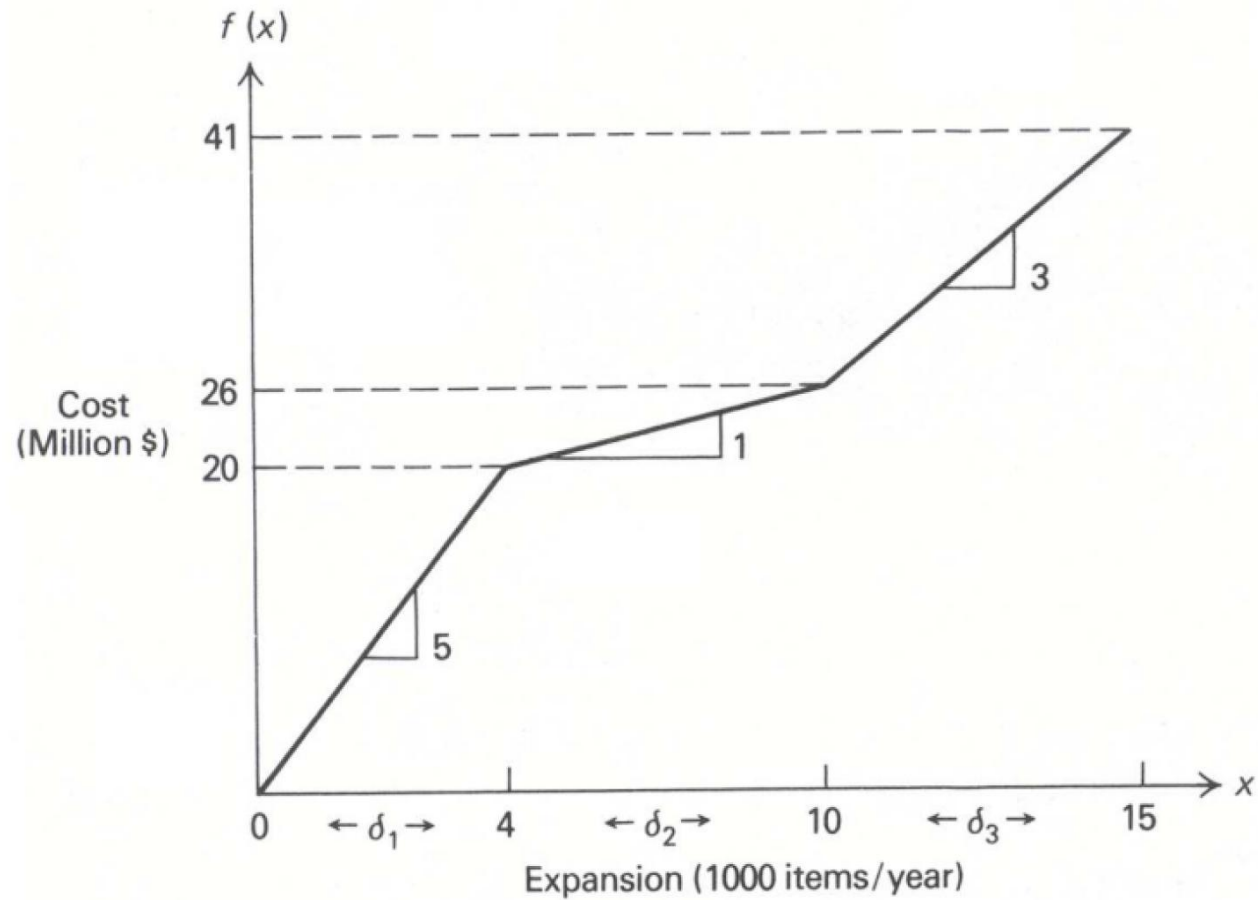
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- Van 2: $x_{A,2} + x_{B,2} + x_{C,2} \leq 80$

3. If there are appointments, the van should be assigned to that school

- School 1 - Van 1: $x_{A,1} \leq 30 \cdot y_{A,1}$
- School 1 - Van 2: $x_{A,2} \leq 80 \cdot y_{A,2}$
- School 2 - Van 1: $x_{B,1} \leq 30 \cdot y_{B,1}$
- School 2 - Van 2: $x_{B,2} \leq 80 \cdot y_{B,2}$
- School 3 - Van 1: $x_{C,1} \leq 30 \cdot y_{C,1}$
- School 3 - Van 2: $x_{C,2} \leq 80 \cdot y_{C,2}$

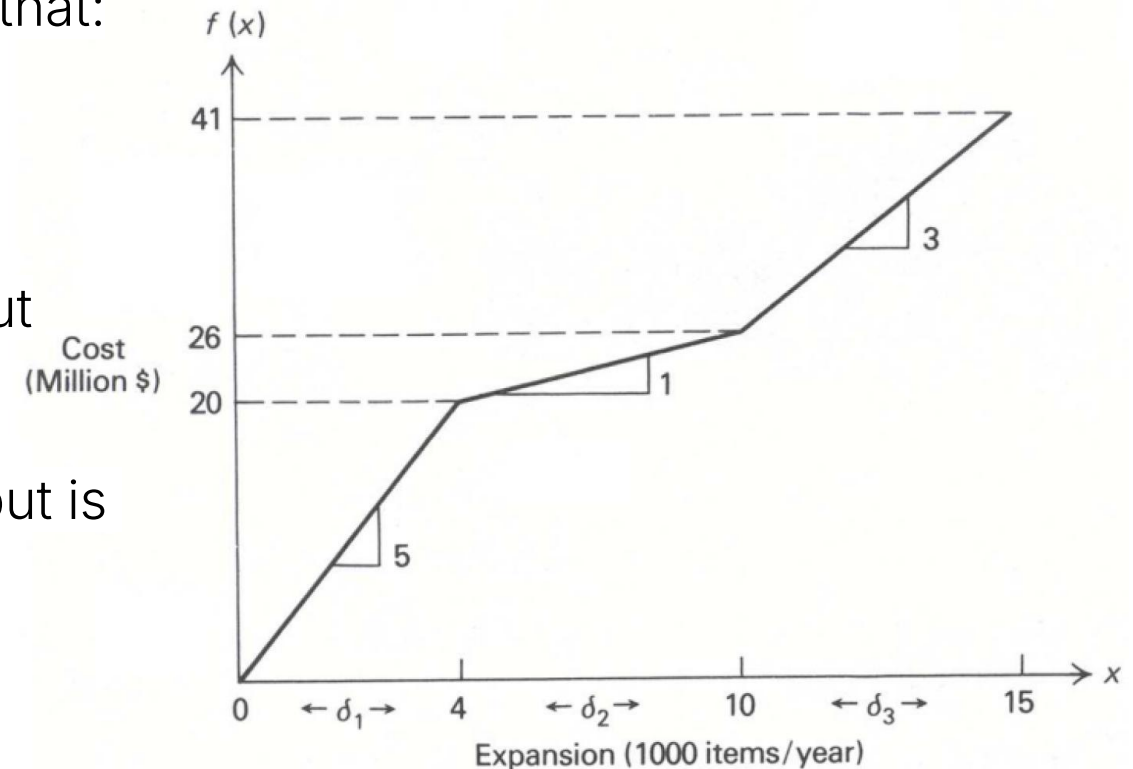
Modeling Piecewise Costs



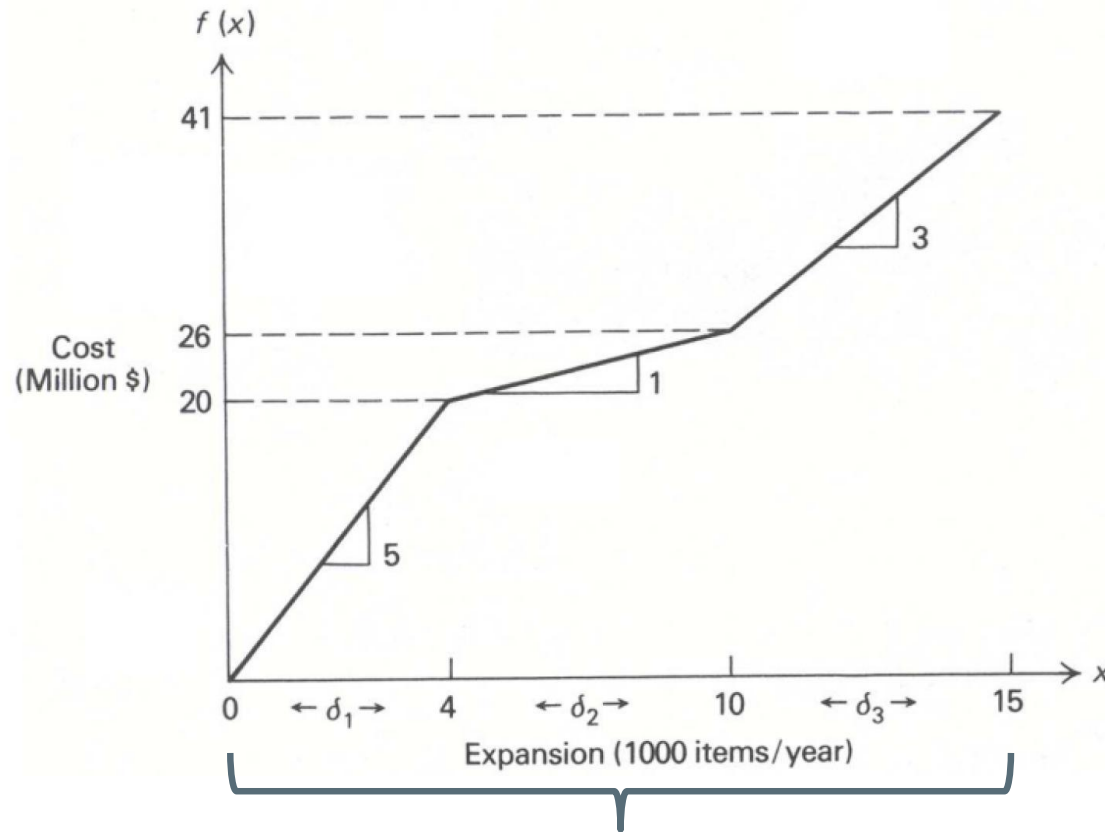
Modeling Piecewise Variable Costs

Define integral variables δ_1 , δ_2 , and δ_3 , so that:

- δ_1 corresponds to the amount by which x exceeds 0, but is less than or equal to 4;
- δ_2 is the amount by which x exceeds 4, but is less than or equal to 10; and
- δ_3 is the amount by which x exceeds 10, but is less than or equal to 15



Modeling Piecewise Costs



where: $x = \delta_1 + \delta_2 + \delta_3$

$$0 \leq \delta_1 \leq 4,$$

$$0 \leq \delta_2 \leq 6,$$

$$0 \leq \delta_3 \leq 5$$

and the total variable cost is given by:

$$\text{Cost} = 5\delta_1 + \delta_2 + 3\delta_3$$

Modeling Piecewise Costs

However, we should have the following conditional constraints:

$$\begin{aligned}\delta_1 &= 4, & \text{if } \delta_2 > 0 \\ \delta_2 &= 6, & \text{if } \delta_3 > 0\end{aligned}$$

Hence, $x = \delta_1 + \delta_2 + \delta_3$

where:

$$\begin{aligned}0 &\leq \delta_1 \leq 4, \\ 0 &\leq \delta_2 \leq 6, \\ 0 &\leq \delta_3 \leq 5\end{aligned}$$

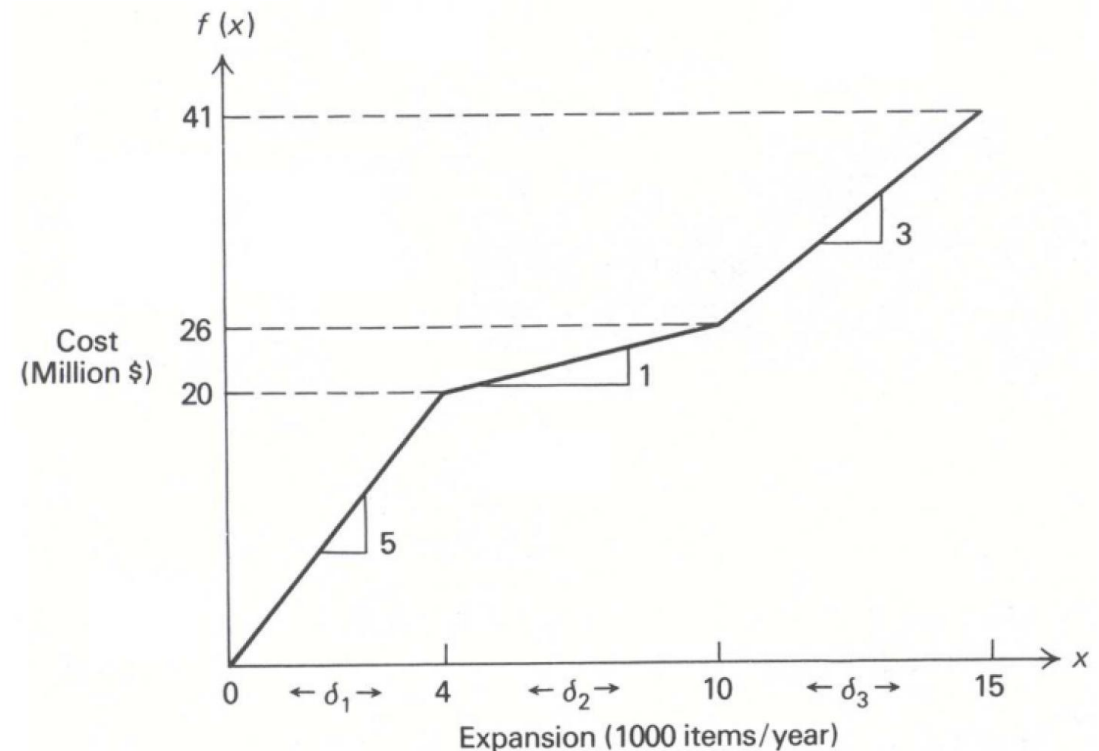
and the **total variable cost** is given by:

$$\text{Cost} = 5\delta_1 + \delta_2 + 3\delta_3$$

Modeling Piecewise Costs

Define $w_1 = \begin{cases} 1 & \text{if } \delta_1 \text{ is at its upper bound} \\ 0 & \text{otherwise} \end{cases}$

$w_2 = \begin{cases} 1 & \text{if } \delta_2 \text{ is at its upper bound} \\ 0 & \text{otherwise} \end{cases}$



Modeling Piecewise Costs

Define

$$w_1 = \begin{cases} 1 & \text{if } \delta_1 \text{ is at its upper bound} \\ 0 & \text{otherwise} \end{cases}$$
$$w_2 = \begin{cases} 1 & \text{if } \delta_2 \text{ is at its upper bound} \\ 0 & \text{otherwise} \end{cases}$$

Then these constraints can be represented as:

$$0 \leq \delta_1 \leq 4, \quad 0 \leq \delta_2 \leq 6, \quad 0 \leq \delta_3 \leq 5$$

$$\begin{aligned} \delta_1 &= 4, & \text{if } \delta_2 > 0 \\ \delta_2 &= 6, & \text{if } \delta_3 > 0 \end{aligned}$$

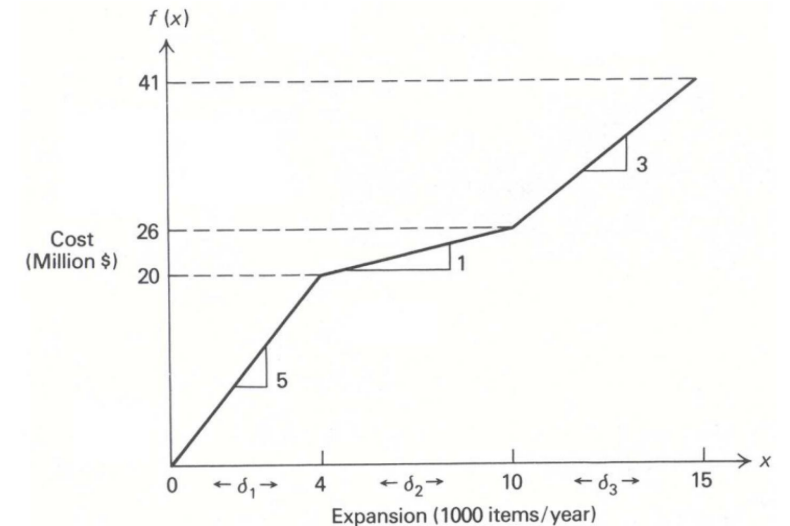
Modeling Piecewise Costs

Define

$$w_1 = \begin{cases} 1 & \text{if } \delta_1 \text{ is at its upper bound} \\ 0 & \text{otherwise} \end{cases}$$

$$w_2 = \begin{cases} 1 & \text{if } \delta_2 \text{ is at its upper bound} \\ 0 & \text{otherwise} \end{cases}$$

$w_1 = 0, w_2 = 0$	corresponding to $0 \leq x \leq 4$
$w_1 = 1, w_2 = 0$	corresponding to $4 \leq x \leq 10$
$w_1 = 1, w_2 = 1$	corresponding to $10 \leq x \leq 15$



Then these constraints can be represented as:

$$0 \leq \delta_1 \leq 4, \quad 0 \leq \delta_2 \leq 6, \quad 0 \leq \delta_3 \leq 5$$

$$\begin{aligned} \delta_1 &= 4, & \text{if } \delta_2 > 0 \\ \delta_2 &= 6, & \text{if } \delta_3 > 0 \end{aligned}$$

$$\begin{aligned} 4w_1 &\leq \delta_1 \leq 4, \\ 6w_2 &\leq \delta_2 \leq 6w_1, \\ 0 &\leq \delta_3 \leq 5w_2 \\ w_1 \text{ and } w_2 &\text{ binary.} \end{aligned}$$